

Smoothing

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Maximum Likelihood Estimation

<s> I am Sam </s>

<s> Sam I am </s>

<s> I do not like green eggs and ham </s>

$$P(w_i \mid w_{i-1}) = \frac{c(w_{i-1}, w_i)}{c(w_{i-1})}$$

$$P(\text{I} \mid \text{<s>}) = \frac{2}{3} = .67$$

$$P(\text{Sam} \mid \text{<s>}) = \frac{1}{3} = .33$$

$$P(\text{am} \mid \text{I}) = \frac{2}{3} = .67$$

$$P(\text{</s>} \mid \text{Sam}) = \frac{1}{2} = 0.5$$

$$P(\text{Sam} \mid \text{am}) = \frac{1}{2} = .5$$

$$P(\text{do} \mid \text{I}) = \frac{1}{3} = .33$$

What kinds of knowledge do N-grams represent?

- $P(\text{to}|\text{want}) = .66$
- $P(\text{want} | \text{spend}) = 0$
- $P(\text{of} | \text{to}) = 0$

Knowledge of grammar

("want" is followed by infinitive "to")

("of" is not a verb)

- $P(\text{dinner}|\text{lunch or}) = .83$
- $P(\text{dinner}|\text{for}) \sim P(\text{lunch}|\text{for})$

Knowledge of meaning

The words "dinner" or "lunch" are semantically related.

- $P(\text{chinese}|\text{want}) > P(\text{english}|\text{want})$

Knowledge about the world

Chinese food is very popular

What about the unknown?

- Data: Tokens=818017, Vocab=21268 (Screen play of TBBT)
- my geology book is missing. With bigram model is,

Dataset Stats: N=818017, V=21268

Step (Token Context)	MLE Calculation	Add-1 Calculation
P('my')	5082/818017	$(5082+1)/(818017+21268)$
P('geology' 'my')	0/5082	$(0+1)/(5082+21268)$
P('book' 'geology')	2/21	$(2+1)/(21+21268)$
P('is' 'book')	0/263	$(0+1)/(263+21268)$
P('missing' 'is')	1/6004	$(1+1)/(6004+21268)$

FINAL MLE PROBABILITY: 0.0

So what can we do?

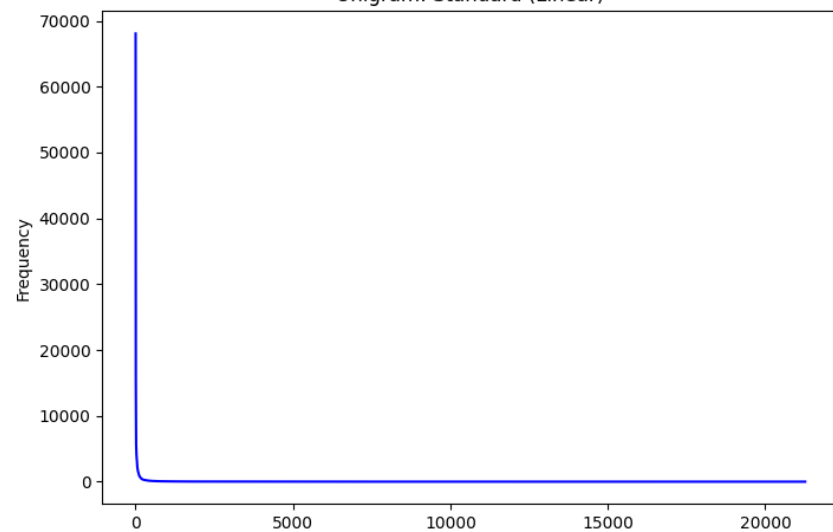
- Add more data? Sure but some words by very nature of their usage are obscure, like obscure. Why?



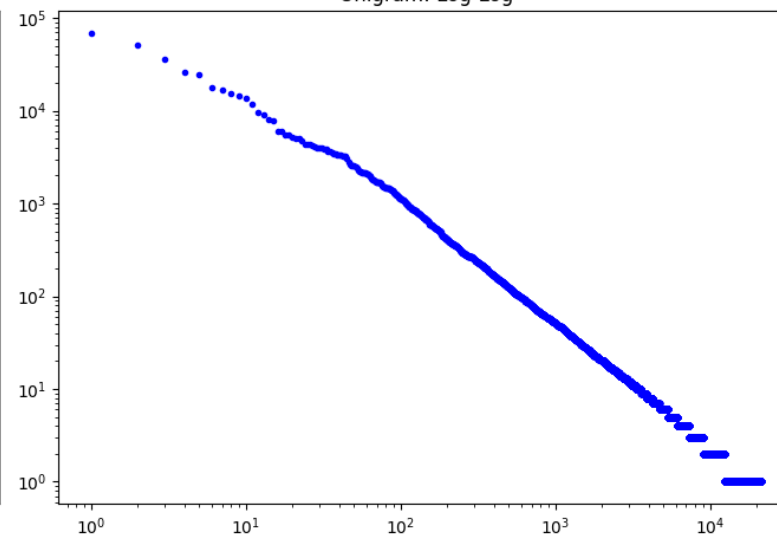
Zipf's Curve

	Unigram				
Top 5	(', 68069)	(', 50547)	('', 36060)	('i', 26236)	('you', 24485)
Top 25% edge	('badum', 6)	('annoyed', 6)	('obi', 6)	('earn', 6)	('almanac', 6)
Middle 5	('farfetched', 2)	('enforce', 2)	('tuvan', 2)	('benedict', 2)	('rupert', 2)
Bottom 5	('guinness', 1)	('skyped', 1)	('corresponded', 1)	('nickels', 1)	('highlighted', 1)
	Bigram				
Top 5	(('', 's'), 13793)	(('', 't'), 7937)	((', 'i'), 7793)	((i, ''), 7394)	((', 'i'), 6931)
Top 25% edge	((', 'cold'), 2)	((a, 'dogapus'), 2)	((dog, 'and'), 2)	((planned, 'on'), 2)	((we, 'owe'), 2)
Middle 5	((and, 'pyramid'), 1)	((pyramid, 'power'), 1)	((power, 'and'), 1)	((and, 'healing'), 1)	((healing, 'crystals'), 1)
Bottom 5	((us, 'sandwiches'), 1)	((would, 'raise'), 1)	((re, 'colleagues'), 1)	((a, 'relation'), 1)	((relation, '...'), 1)

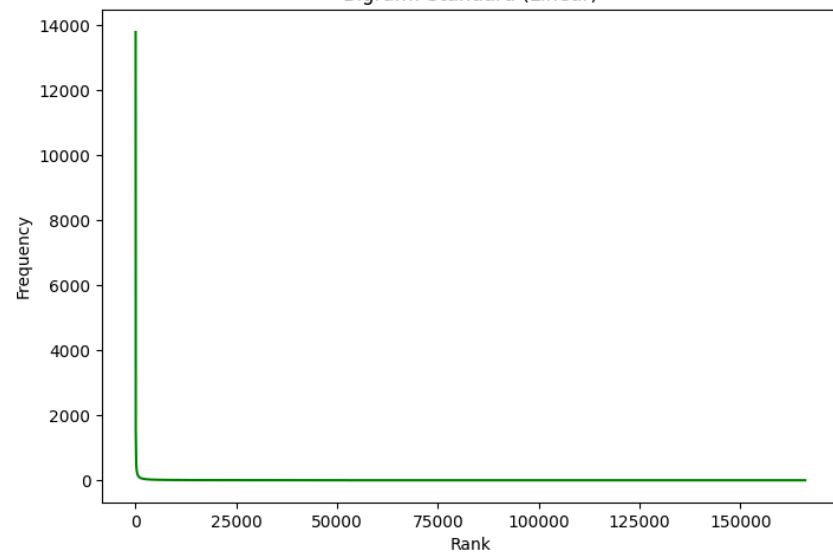
Unigram: Standard (Linear)



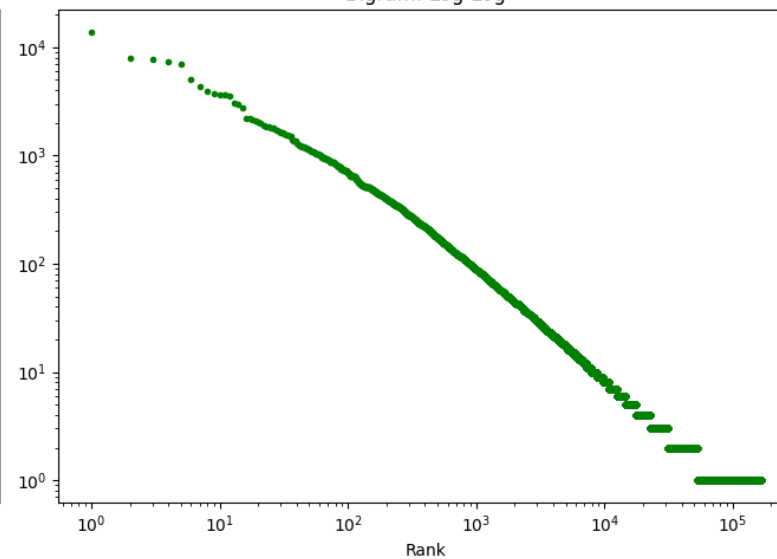
Unigram: Log-Log



Bigram: Standard (Linear)



Bigram: Log-Log



So what can we do?

- Add more data? Sure but some words by very nature of their usage are obscure, like obscure. Why?
- Something simple to increase the count a bit?

Add-1 (Laplace) Smoothing

$$P_{Laplace}(w_i) = \frac{C(w_i)+1}{N+V}$$

$$P_{Laplace}(w_i | w_{i-1}) = \frac{C(w_{i-1}, w_i) + 1}{C(w_{i-1}) + V}$$

V is the vocabulary size and N is the total token count

P('my') -> MLE: 0.0062, Add-1: 0.0061

P('spot') -> MLE: 0.0001, Add-1: 0.0001

P('spot'|'my') -> **MLE: 0.0069, Add-1: 0.0014**

P('geology'|'my') -> MLE: 0, Add-1: 0.00003795

- Too much of probability “mass” is shifted.
- From frequently seen to unknown words.

Dataset Stats: N=818017, V=21268

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FINAL MLE PROBABILITY: 0.0

FINAL ADD-1 PROBABILITY: 1.1031760585e-19



Discounted Count

- So how many times would we have to see ... to see this
“**smoothed over**” **probability**? => Discounted count.

Unigram $c^* = (c + 1) \frac{N}{N+V}$

Bigram $c^*(w_{i-1}, w_i) = [C(w_{i-1}, w_i) + 1] \frac{C(w_{i-1})}{C(w_{i-1})+V}$

Token	Raw Count	Add-1 Prob	Discounted (c*)
my	5082	0.0061	4954.19
spot	121	0.0001	118.91
my spot (bi)	35	0.0014	6.94

So what can we do?

- Add more data? Sure but some words by very nature of their usage are obscure, like obscure. Why?
- Something simple to increase the count a bit?
- Look at smaller models?

Backoff to smaller models

- Prioritize the highest-order model (say 4-gram) and **back-off** to lower models (trigram, bigram and unigram) if current count is zero. Or,

$$P_{backoff}(w_i | w_{i-1}) = \begin{cases} \frac{C^*(w_{i-1}, w_i)}{C(w_{i-1})} & \text{if } C(w_{i-1}, w_i) > 0 \\ \alpha(w_{i-1}) \cdot P(w_i) & \text{if } C(w_{i-1}, w_i) = 0 \end{cases}$$

C^* is discounted count and α is normalization factor. Because you won't just have probability of bigram (which sums to 1) but will also introduce unigram probability while finding probability of sentences.

Interpolate multiple models to act as one

$$P_{interp}(w_i \mid w_{i-1}) = \lambda_2 \cdot P_{MLE}(w_i \mid w_{i-1}) + \lambda_1 \cdot P(w_i) \quad \lambda_2 + \lambda_1 = 1$$

- How to calculate λ_2, λ_1 ? Held out data



So what can we do?

- Add more data? Sure but some words by very nature of their usage are obscure, like obscure. Why?
- Something simple to increase the count a bit?
- Look at smaller models?
- Look at how words in similar (almost unknown like) state behave, may be something is there?

San _____

- Fransico . What can be before it?
- Most likely San. Hence, bigram count of San Fransico will be high.
- Fransico most likely will follow only San, but its unigram can be high. So, e.g. I can't see without my reading XXXXXX
 - Assume nothing is there after reading in **data**. But has lot of San Fransico.
 - So backoff will give unigram Fransico very high probability. But,
 - It only appears with San and any other word like glasses would be more, diverse and used in different context.

Don't ask what kind of dataset this is.

Again, don't ask about the dataset.

Kneser-Ney Intuition

- Absolute Discounting: take small probability mass from all the words and save it for unknown words.
- Continuation probability:
 - Instead of checking how often **glasses** appears, we check how many different words it appear with.

Kneser-Ney Smoothing

- Continuation Probability:

$$P_{\text{continuation}}(w_i) = \frac{|\{w_{i-1} : c(w_{i-1}w_i) > 0\}|}{\sum_{w_i} |\{w_{i-1} : c(w_{i-1}w_i) > 0\}|}$$

- Kneser-Ney backoff is formalized as follows assuming proper coefficient α on the backoff to make everything sum to one:

$$P_{KN}(w_i | w_{i-1}) = \begin{cases} \frac{c(w_{i-1}w_i) - D}{c(w_{i-1})}, & \text{if } c(w_{i-1}w_i) > 0 \\ \alpha(w_i) \frac{|\{w_{i-1} : c(w_{i-1}w_i) > 0\}|}{\sum_{w_i} |\{w_{i-1} : c(w_{i-1}w_i) > 0\}|}, & \text{otherwise} \end{cases}$$

Interpolated vs Backoff form of Kneser-Ney

- Kneser-Ney **backoff** algorithm was shown to be less superior to its **interpolated** version.
- **Interpolated Kneser-Ney** discounting can be computed with an equation like the following (omitting the computation of β):

$$P_{KN}(w_i | w_{i-1}) = \frac{c(w_{i-1}w_i) - D}{c(w_{i-1})} + \beta(w_i) \frac{|\{w_{i-1} : c(w_{i-1}w_i) > 0\}|}{\sum_{w_i} |\{w_{i-1} : c(w_{i-1}w_i) > 0\}|}$$

- Practical note – it turns out that any interpolation model can be represented as a backoff model, and hence stored in ARPA backoff format. The interpolation is done when the model is built, thus the ‘bigram’ probability stored in the backoff format is really ‘bigram already interpolated with unigram’.

Example...

Bigram (w prev)	(count)	KN C*
geology my	(0)	0.00
book geology	(2)	1.25
is book	(0)	0.00
missing is	(1)	0.25

KN C* : Kneser-Ney Discounted Count

Witten-Bell Intuition

- In this specific context, how often do I encounter something I've never seen before? Or,
- If $P(w_i | w_{i-1}, \dots, w_{i-m}) = 0$, then the smoothed probability $P_{WB}(w_i | w_{i-1}, \dots, w_{i-m})$ is higher if the sequence $w_{i-1}, \dots, w_{i-m}$ occurs with many different words w_i
 - **The Scientist (Sheldon) Dialogue 1:** lepton ...
 - **The Scientist (Sheldon) Dialogue 2:** photon
 - **The Scientist (Sheldon) Dialogue 3:** boson
 - **The Scientist (Sheldon) Dialogue 4:** ?
 - How are? More contextual diversity in “Dialogue X:” compared to “how are”

Witten-Bell Smoothing

- Let's consider bi-grams
 - $T(w_{i-1})$ is the number of different words (types) that occur to the right of w_{i-1}
 - $N(w_{i-1})$ is the number of all word occurrences (tokens) to the right of w_{i-1}
 - $Z(w_{i-1})$ is the number of bigrams in the current data set starting with w_{i-1} that do not occur in the training data

Witten-Bell Smoothing

- If $c(w_{i-1}, w_i) = 0$

$$P^{WB}(w_i | w_{i-1}) = \frac{T(w_{i-1})}{Z(w_{i-1})(N + T(w_{i-1}))}$$

- If $c(w_{i-1}, w_i) > 0$

$$P^{WB}(w_i | w_{i-1}) = \frac{c(w_{i-1}w_i)}{N(w_{i-1}) + T(w_{i-1})}$$

Bigram (w prev)	(count)	KN C*	WB C*
geology my	(0)	0.00	0.00
book geology	(2)	1.25	1.40
is book	(0)	0.00	0.00
missing is	(1)	0.25	0.84

Addone smoothing: 1.1031760585e-19

KN Interp Probability: 4.1700358136e-14

KN Backoff Probability: 2.7841245198e-14

WB Interp Probability: 6.5218859977e-14

WB Backoff Probability: 6.1511619997e-14

KN C* : Kneser-Ney Discounted Count

WB C* : Witten-Bell Discounted Count